

Analisis Keterampilan Berpikir Kritis Siswa

Okta

"the curious mind"

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1 METODE

1.1 Analisis Rasch

Analisis dilakukan menggunakan Partial Credit Model (PCM). Pemilihan PCM didasari oleh data jawaban instrumen penelitian yang bersifat politomi dengan rentang skor 0-4 (Masters, 1982). Selain itu, PCM dipilih karena skala nilai tidak sama antar soal dan tingkatan nilai antar kategori respon tidak dianggap sama nilainya (Debelak dkk., 2022; Masters, 1982). Untuk menguji asumsi-asumsi dasar pemodelan Rasch, prosedur-prosedur berikut dilakukan. Pertama, dilakukan uji unidimensionalitas untuk mengukur

2 HASIL

2.1 Validitas dan Reliabilitas Instrumen (Kualitas Soal)

2.1.1 Uji Unidimensionalitas

Martin-Loef-Test (split criterion: mean)

LR-value: 184.706

Chi-square df: 503

p-value: 1

2.1.2 Independensi Lokal (Local Item Independence)

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12
S1	1.00	0.00	0.13	0.10	0.03	-0.02	-0.29	-0.24	-0.21	-0.16	0.12	-0.25
S2	0.00	1.00	-0.09	0.03	0.02	-0.13	-0.17	0.07	0.02	-0.18	-0.25	-0.10
S3	0.13	-0.09	1.00	0.19	0.05	0.08	-0.27	-0.04	-0.01	-0.21	0.06	-0.20
S4	0.10	0.03	0.19	1.00	-0.16	-0.02	-0.16	-0.15	-0.18	-0.05	-0.18	-0.09
S5	0.03	0.02	0.05	-0.16	1.00	-0.07	-0.09	-0.05	-0.12	-0.32	-0.10	-0.26
S6	-0.02	-0.13	0.08	-0.02	-0.07	1.00	-0.08	-0.13	-0.25	-0.14	0.07	-0.11

S7	-0.29	-0.17	-0.27	-0.16	-0.09	-0.08	1.00	-0.05	-0.16	-0.15	-0.39	-0.07
S8	-0.24	0.07	-0.04	-0.15	-0.05	-0.13	-0.05	1.00	0.04	-0.06	-0.09	-0.17
S9	-0.21	0.02	-0.01	-0.18	-0.12	-0.25	-0.16	0.04	1.00	-0.03	0.05	0.19
S10	-0.16	-0.18	-0.21	-0.05	-0.32	-0.14	-0.15	-0.06	-0.03	1.00	0.03	0.14
S11	0.12	-0.25	0.06	-0.18	-0.10	0.07	-0.39	-0.09	0.05	0.03	1.00	-0.01
S12	-0.25	-0.10	-0.20	-0.09	-0.26	-0.11	-0.07	-0.17	0.19	0.14	-0.01	1.00

2.1.3 Item Fit (Kesesuaian Butir Soal)

Itemfit Statistics:

	Chisq	df	p-value	Outfit MSQ	Infit MSQ	Outfit t	Infit t	Discrim
S1	76.388	101	0.968	0.749	0.785	-1.909	-2.282	0.606
S2	93.314	101	0.694	0.915	0.962	-0.521	-0.204	0.249
S3	90.663	101	0.760	0.889	0.914	-0.944	-0.724	0.432
S4	101.708	101	0.462	0.997	0.994	0.018	-0.014	0.276
S5	108.038	101	0.298	1.059	1.050	0.517	0.458	0.215
S6	96.788	101	0.600	0.949	0.916	-0.258	-0.538	0.347
S7	179.973	101	0.000	1.764	1.429	3.980	3.445	-0.041
S8	100.127	101	0.506	0.982	0.994	-0.032	0.030	0.106
S9	75.659	101	0.972	0.742	0.778	-1.618	-1.389	0.529
S10	82.940	101	0.905	0.813	0.800	-1.438	-1.809	0.530
S11	73.544	101	0.982	0.721	0.724	-2.576	-2.589	0.665
S12	88.530	101	0.808	0.868	0.877	-0.892	-0.955	0.422

2.1.4 Item Reliability & Separation Index

Item Reliability (Separation) : 0.905

2.1.5 Fungsi Kategori Skor (Threshold/Category Functioning)

Design Matrix Block 1:

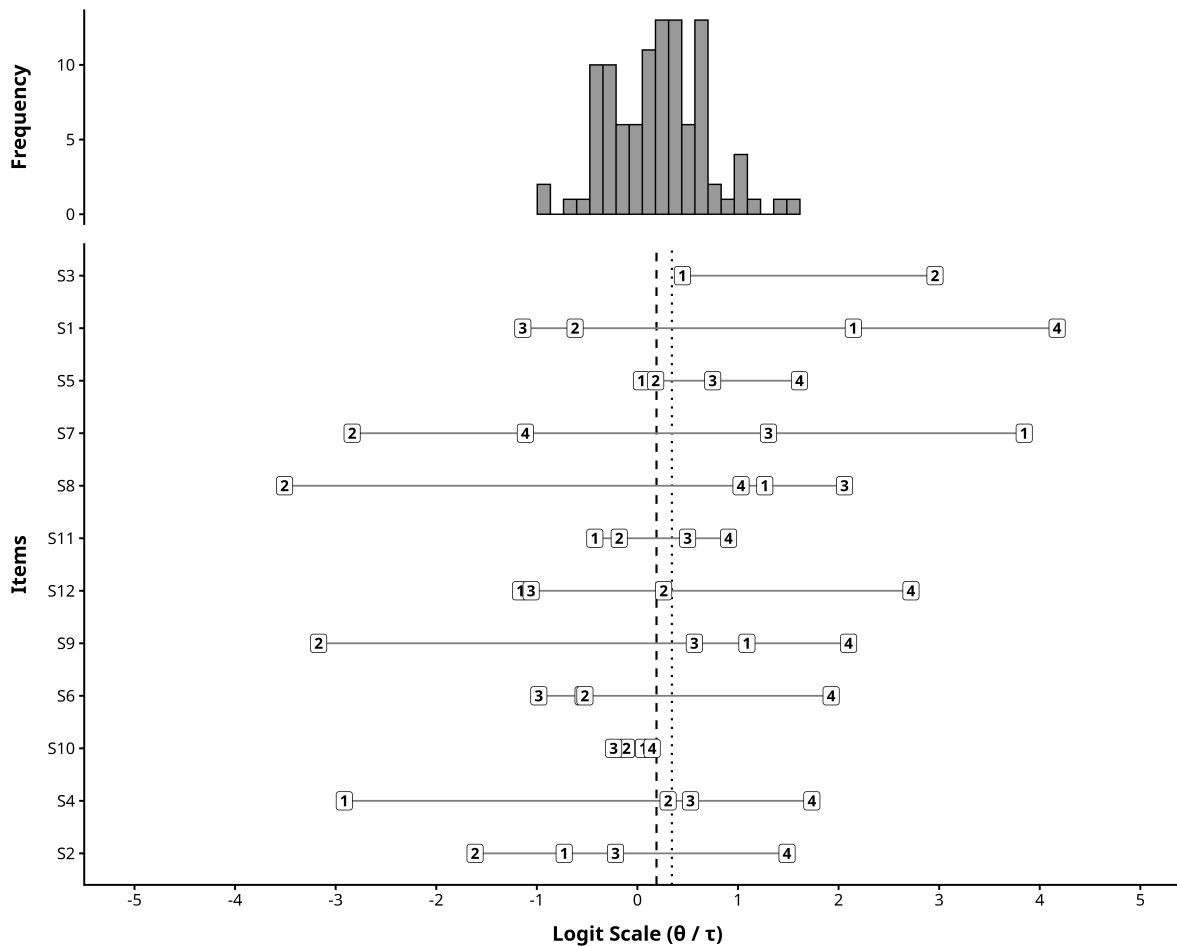
	Location	Threshold 1	Threshold 2	Threshold 3	Threshold 4
S1	1.13996	2.14635	-0.61988	-1.13939	4.17276
S2	-0.26814	-0.72817	-1.61408	-0.21779	1.48748
S3	1.70287	0.44718	2.95857	NA	NA
S4	-0.08668	-2.91431	0.30488	0.52865	1.73406
S5	0.64538	0.04046	0.18169	0.74833	1.61105
S6	-0.02914	-0.54046	-0.52178	-0.98036	1.92603
S7	0.29946	3.84541	-2.83638	1.30301	-1.11422
S8	0.21198	1.26632	-3.50799	2.05951	1.03006
S9	0.14619	1.08858	-3.16922	0.56570	2.09971
S10	-0.03455	0.05970	-0.10812	-0.23645	0.14665
S11	0.20022	-0.42339	-0.18102	0.49899	0.90629
S12	0.19040	-1.16204	0.26086	-1.05675	2.71951

2.2 Profil Kemampuan Siswa

2.2.1 Person Reliability & Separation Index

Person Reliability (Separation) : 0.62

2.2.2 Person-Item Map (*Wright Map*)



Gambar 1: Treshold Map

2.2.3 Person Measure (Skor Logit Kemampuan Siswa)

Estimation of Ability Parameters

Collapsed log-likelihood: -333.3505

Number of iterations: 8

Number of parameters: 25

ML estimated ability parameters (without spline interpolated values):

	Estimate	Std. Err.	2.5 %	97.5 %
theta P1	0.97725925	0.3424443	0.306080736	1.64843777
theta P2	0.57895229	0.2925880	0.005490324	1.15241425
theta P3	0.26784732	0.2690112	-0.259404847	0.79509949
theta P4	0.12552576	0.2652786	-0.394410661	0.64546218
theta P5	0.34115479	0.2727220	-0.193370521	0.87568010
theta P6	0.19619099	0.2665656	-0.326268037	0.71865001
theta P7	-0.38571501	0.2816907	-0.937818696	0.16638867
theta P8	-0.01515017	0.2657973	-0.536103287	0.50580295
theta P9	0.57895229	0.2925880	0.005490324	1.15241425
theta P10	0.19619099	0.2665656	-0.326268037	0.71865001
theta P11	0.66739248	0.3024696	0.074563048	1.26022192
theta P12	0.34115479	0.2727220	-0.193370521	0.87568010
theta P13	0.05525876	0.2650517	-0.464233038	0.57475055
theta P14	0.41685831	0.2778122	-0.127643636	0.96136026
theta P15	-0.15834807	0.2698998	-0.687341890	0.37064575
theta P16	0.76234696	0.3140905	0.146740952	1.37795298
theta P17	0.34115479	0.2727220	-0.193370521	0.87568010
theta P18	-0.01515017	0.2657973	-0.536103287	0.50580295
theta P19	0.34115479	0.2727220	-0.193370521	0.87568010
theta P20	0.57895229	0.2925880	0.005490324	1.15241425
theta P21	0.12552576	0.2652786	-0.394410661	0.64546218
theta P22	0.34115479	0.2727220	-0.193370521	0.87568010
theta P23	0.76234696	0.3140905	0.146740952	1.37795298
theta P24	0.19619099	0.2665656	-0.326268037	0.71865001
theta P25	0.97725925	0.3424443	0.306080736	1.64843777
theta P26	0.12552576	0.2652786	-0.394410661	0.64546218
theta P27	0.05525876	0.2650517	-0.464233038	0.57475055
theta P28	-0.08619818	0.2674366	-0.610364200	0.43796785
theta P29	0.41685831	0.2778122	-0.127643636	0.96136026
theta P30	-0.01515017	0.2657973	-0.536103287	0.50580295
theta P31	1.55068016	0.4178717	0.731666638	2.36969369
theta P32	1.38502884	0.3964940	0.607914947	2.16214274
theta P33	0.57895229	0.2925880	0.005490324	1.15241425
theta P34	0.57895229	0.2925880	0.005490324	1.15241425
theta P35	0.49580388	0.2843982	-0.061606270	1.05321403
theta P36	-0.46653497	0.2869777	-1.029000860	0.09593092
theta P37	0.19619099	0.2665656	-0.326268037	0.71865001
theta P38	0.12552576	0.2652786	-0.394410661	0.64546218
theta P39	-0.23203760	0.2731272	-0.767357108	0.30328192
theta P40	-0.15834807	0.2698998	-0.687341890	0.37064575
theta P41	-0.23203760	0.2731272	-0.767357108	0.30328192
theta P42	-0.30768845	0.2770691	-0.850733952	0.23535705
theta P43	-0.23203760	0.2731272	-0.767357108	0.30328192
theta P44	0.12552576	0.2652786	-0.394410661	0.64546218
theta P45	0.66739248	0.3024696	0.074563048	1.26022192

theta P46	0.12552576	0.2652786	-0.394410661	0.64546218
theta P47	-0.08619818	0.2674366	-0.610364200	0.43796785
theta P48	0.49580388	0.2843982	-0.061606270	1.05321403
theta P49	0.41685831	0.2778122	-0.127643636	0.96136026
theta P50	-0.01515017	0.2657973	-0.536103287	0.50580295
theta P51	0.97725925	0.3424443	0.306080736	1.64843777
theta P52	0.57895229	0.2925880	0.005490324	1.15241425
theta P53	0.66739248	0.3024696	0.074563048	1.26022192
theta P54	0.26784732	0.2690112	-0.259404847	0.79509949
theta P55	0.66739248	0.3024696	0.074563048	1.26022192
theta P56	-0.55058571	0.2929460	-1.124749263	0.02357784
theta P57	0.19619099	0.2665656	-0.326268037	0.71865001
theta P58	-0.15834807	0.2698998	-0.687341890	0.37064575
theta P59	0.41685831	0.2778122	-0.127643636	0.96136026
theta P60	0.49580388	0.2843982	-0.061606270	1.05321403
theta P61	-0.30768845	0.2770691	-0.850733952	0.23535705
theta P62	0.12552576	0.2652786	-0.394410661	0.64546218
theta P63	-0.30768845	0.2770691	-0.850733952	0.23535705
theta P64	0.19619099	0.2665656	-0.326268037	0.71865001
theta P65	0.49580388	0.2843982	-0.061606270	1.05321403
theta P66	0.12552576	0.2652786	-0.394410661	0.64546218
theta P67	-0.38571501	0.2816907	-0.937818696	0.16638867
theta P68	0.97725925	0.3424443	0.306080736	1.64843777
theta P69	0.41685831	0.2778122	-0.127643636	0.96136026
theta P70	0.66739248	0.3024696	0.074563048	1.26022192
theta P71	0.19619099	0.2665656	-0.326268037	0.71865001
theta P72	-0.23203760	0.2731272	-0.767357108	0.30328192
theta P73	0.57895229	0.2925880	0.005490324	1.15241425
theta P74	-0.93036312	0.3260111	-1.569333209	-0.29139304
theta P75	-0.93036312	0.3260111	-1.569333209	-0.29139304
theta P76	-0.38571501	0.2816907	-0.937818696	0.16638867
theta P77	-0.01515017	0.2657973	-0.536103287	0.50580295
theta P78	0.66739248	0.3024696	0.074563048	1.26022192
theta P79	0.49580388	0.2843982	-0.061606270	1.05321403
theta P80	-0.23203760	0.2731272	-0.767357108	0.30328192
theta P81	-0.46653497	0.2869777	-1.029000860	0.09593092
theta P82	0.26784732	0.2690112	-0.259404847	0.79509949
theta P83	0.26784732	0.2690112	-0.259404847	0.79509949
theta P84	-0.46653497	0.2869777	-1.029000860	0.09593092
theta P85	-0.63834965	0.2996574	-1.225667284	-0.05103201
theta P86	-0.01515017	0.2657973	-0.536103287	0.50580295
theta P87	0.34115479	0.2727220	-0.193370521	0.87568010
theta P88	-0.46653497	0.2869777	-1.029000860	0.09593092
theta P89	0.34115479	0.2727220	-0.193370521	0.87568010
theta P90	1.10017273	0.3589876	0.396569907	1.80377556
theta P91	0.49580388	0.2843982	-0.061606270	1.05321403
theta P92	-0.15834807	0.2698998	-0.687341890	0.37064575
theta P93	-0.38571501	0.2816907	-0.937818696	0.16638867

theta P94	0.86515729	0.3274403	0.223386166	1.50692841
theta P95	-0.30768845	0.2770691	-0.850733952	0.23535705
theta P96	0.05525876	0.2650517	-0.464233038	0.57475055
theta P97	-0.38571501	0.2816907	-0.937818696	0.16638867
theta P98	0.34115479	0.2727220	-0.193370521	0.87568010
theta P99	0.26784732	0.2690112	-0.259404847	0.79509949
theta P100	-0.46653497	0.2869777	-1.029000860	0.09593092
theta P101	0.19619099	0.2665656	-0.326268037	0.71865001
theta P102	-0.30768845	0.2770691	-0.850733952	0.23535705

Daftar Pustaka

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